
FRACTALS

11.0 Introduction

Back in Chapter 9, we found that the solutions of the Lorenz equations settle down to a complicated set in phase space. This set is the strange attractor. As Lorenz (1963) realized, the geometry of this set must be very peculiar, something like an “infinite complex of surfaces.” In this chapter we develop the ideas needed to describe such strange sets more precisely. The tools come from fractal geometry.

Roughly speaking, *fractals* are complex geometric shapes with fine structure at arbitrarily small scales. Usually they have some degree of self-similarity. In other words, if we magnify a tiny part of a fractal, we will see features reminiscent of the whole. Sometimes the similarity is exact; more often it is only approximate or statistical.

Fractals are of great interest because of their exquisite combination of beauty, complexity, and endless structure. They are reminiscent of natural objects like mountains, clouds, coastlines, blood vessel networks, and even broccoli, in a way that classical shapes like cones and squares can’t match. They have also turned out to be useful in scientific applications ranging from computer graphics and image compression to the structural mechanics of cracks and the fluid mechanics of viscous fingering.

Our goals in this chapter are modest. We want to become familiar with the simplest fractals and to understand the various notions of fractal dimension. These ideas will be used in Chapter 12 to clarify the geometric structure of strange attractors.

Unfortunately, we will not be able to delve into the scientific applications of fractals, nor the lovely mathematical theory behind them. For the clearest introduction to the theory and applications of fractals, see Falconer (1990). The books of Mandelbrot (1982), Peitgen and Richter (1986), Barnsley (1988), Feder (1988), and Schroeder (1991) are also recommended for their many fascinating pictures and examples.

11.1 Countable and Uncountable Sets

This section reviews the parts of set theory that we'll need in later discussions of fractals. You may be familiar with this material already; if not, read on.

Are some infinities larger than others? Surprisingly, the answer is yes. In the late 1800s, Georg Cantor invented a clever way to compare different infinite sets. Two sets X and Y are said to have the same *cardinality* (or number of elements) if there is an invertible mapping that pairs each element $x \in X$ with precisely one $y \in Y$. Such a mapping is called a **one-to-one correspondence**; it's like a buddy system, where every x has a buddy y , and no one in either set is left out or counted twice.

A familiar infinite set is the set of natural numbers $\mathbf{N} = \{1, 2, 3, 4, \dots\}$. This set provides a basis for comparison—if another set X can be put into one-to-one correspondence with the natural numbers, then X is said to be **countable**. Otherwise X is **uncountable**.

These definitions lead to some surprising conclusions, as the following examples show.

EXAMPLE 11.1.1:

Show that the set of even natural numbers $E = \{2, 4, 6, \dots\}$ is countable.

Solution: We need to find a one-to-one correspondence between E and $\mathbf{N}$. Such a correspondence is given by the invertible mapping that pairs each natural number n with the even number $2n$; thus $1 \leftrightarrow 2$, $2 \leftrightarrow 4$, $3 \leftrightarrow 6$, and so on.

Hence there are exactly as many even numbers as natural numbers. You might have thought that there would be only *half as* many, since all the odd numbers are missing! ■

There is an equivalent characterization of countable sets which is frequently useful. A set X is countable if it can be written as a list $\{x_1, x_2, x_3, \dots\}$, with every $x \in X$ appearing somewhere in the list. In other words, given any x , there is some finite n such that $x_n = x$.

A convenient way to exhibit such a list is to give an algorithm that systematically counts the elements of X . This strategy is used in the next two examples.

EXAMPLE 11.1.2:

Show that the integers are countable.

Solution: Here's an algorithm for listing all the integers: We start with 0 and then work in order of increasing absolute value. Thus the list is $\{0, 1, -1, 2, -2, 3, -3, \dots\}$. Any particular integer appears eventually, so the integers are countable. ■

EXAMPLE 11.1.3:

Show that the positive rational numbers are countable.

Solution: Here's a *wrong* way: we start listing the numbers $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \dots$ in order. Unfortunately we never finish the $\frac{1}{n}$'s and so numbers like $\frac{2}{3}$ are never counted!

The right way is to make a table where the pq -th entry is p/q . Then the rationals can be counted by the weaving procedure shown in Figure 11.1.1. Any given p/q is reached after a finite number of steps, so the rationals are countable. ■

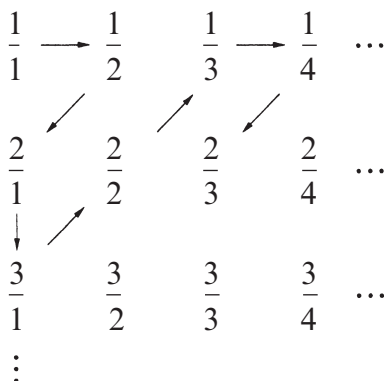


Figure 11.1.1

Now we consider our first example of an uncountable set.

EXAMPLE 11.1.4:

Let X denote the set of all real numbers between 0 and 1. Show that X is uncountable.

Solution: The proof is by contradiction. If X were countable, we could list all the real numbers between 0 and 1 as a set $\{x_1, x_2, x_3, \dots\}$. Rewrite these numbers in decimal form:

$$\begin{aligned}x_1 &= 0.x_{11}x_{12}x_{13}x_{14}\dots \\x_2 &= 0.x_{21}x_{22}x_{23}x_{24}\dots \\x_3 &= 0.x_{31}x_{32}x_{33}x_{34}\dots \\&\vdots\end{aligned}$$

where x_{ij} denotes the j th digit of the real number x_i .

To obtain a contradiction, we'll show that there's a number r between 0 and 1 that is *not* on the list. Hence any list is necessarily incomplete, and so the reals are uncountable.

We construct r as follows: its first digit is *anything other than* x_{11} , the first digit of x_1 . Similarly, its second digit is anything other than the second digit of x_2 . In general, the n th digit of r is $\bar{x}_{nm}$, defined as any digit other than x_{nm} . Then we claim that the number $r = \bar{x}_{11}\bar{x}_{22}\bar{x}_{33}\dots$ is not on the list. Why not? It can't be equal to x_1 , because it differs from x_1 in the first decimal place. Similarly, r differs from x_2 in the second decimal place, from x_3 in the third decimal place, and so on. Hence r is not on the list, and thus X is uncountable. ■

This argument (devised by Cantor) is called the **diagonal argument**, because r is constructed by changing the diagonal entries x_{nn} in the matrix of digits $[x_{ij}]$.

11.2 Cantor Set

Now we turn to another of Cantor's creations, a fractal known as the Cantor set. It is simple and therefore pedagogically useful, but it is also much more than that—as we'll see in Chapter 12, the Cantor set is intimately related to the geometry of strange attractors.

Figure 11.2.1 shows how to construct the Cantor set.

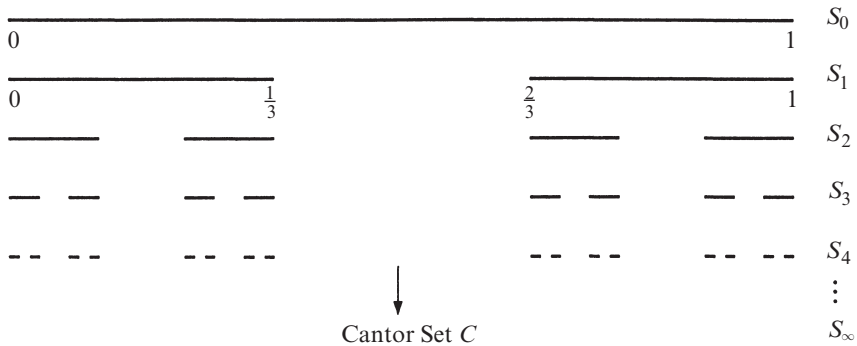


Figure 11.2.1

We start with the closed interval $S_0 = [0, 1]$ and remove its open middle third, i.e., we delete the interval $(\frac{1}{3}, \frac{2}{3})$ and leave the endpoints behind. This produces the pair of closed intervals shown as S_1 . Then we remove the open middle thirds of *those* two intervals to produce S_2 , and so on. The limiting set $C = S_\infty$ is the **Cantor set**. It is difficult to visualize, but Figure 11.2.1 suggests that it consists of an infinite number of infinitesimal pieces, separated by gaps of various sizes.

Fractal Properties of the Cantor Set

The Cantor set C has several properties that are typical of fractals more generally:

1. *C has structure at arbitrarily small scales.* If we enlarge part of C repeatedly, we continue to see a complex pattern of points separated by gaps of various sizes. This structure is neverending, like worlds within worlds. In contrast, when we look at a smooth curve or surface under repeated magnification, the picture becomes more and more featureless.
2. *C is self-similar.* It contains smaller copies of itself at all scales. For instance, if we take the left part of C (the part contained in the interval $[0, \frac{1}{3}]$) and enlarge it by a factor of three, we get C back again. Similarly, the parts of C in each of the four intervals of S_2 are geometrically similar to C , except scaled down by a factor of nine.

If you're having trouble seeing the self-similarity, it may help to think about the sets S_n rather than the mind-boggling set S_∞ . Focus on the left half of S_2 —it looks just like S_1 , except three times smaller. Similarly, the left half of S_3 is S_2 , reduced by a factor of three. In general, the left half of S_{n+1} looks like *all* of S_n , scaled down by three. Now set $n = \infty$. The conclusion is that the left half of S_∞ looks like S_∞ , scaled down by three, just as we claimed earlier.

Warning: The strict self-similarity of the Cantor set is found only in the simplest fractals. More general fractals are only approximately self-similar.

3. *The dimension of C is not an integer.* As we'll show in Section 11.3, its dimension is actually $\ln 2 / \ln 3 \approx 0.63$! The idea of a noninteger dimension is bewildering at first, but it turns out to be a natural generalization of our intuitive ideas about dimension, and provides a very useful tool for quantifying the structure of fractals.

Two other properties of the Cantor set are worth noting, although they are not fractal properties as such: *C has measure zero* and *it consists of uncountably many points*. These properties are clarified in the examples below.

EXAMPLE 11.2.1:

Show that the *measure* of the Cantor set is zero, in the sense that it can be covered by intervals whose total length is arbitrarily small.

Solution: Figure 11.2.1 shows that each set S_n completely covers all the sets that come after it in the construction. Hence the Cantor set $C = S_\infty$ is covered by *each* of the sets S_n . So the total length of the Cantor set must be less than the total length of S_n , for any n . Let L_n denote the length of S_n . Then from Figure 11.2.1 we see that $L_0 = 1$, $L_1 = \frac{2}{3}$, $L_2 = (\frac{2}{3})(\frac{2}{3}) = (\frac{2}{3})^2$, and in general, $L_n = (\frac{2}{3})^n$. Since $L_n \rightarrow 0$ as $n \rightarrow \infty$, the Cantor set has a total length of zero. ■

Example 11.2.1 suggests that the Cantor set is “small” in some sense. On the other hand, it contains tremendously many points—uncountably many, in fact. To see this, we first develop an elegant characterization of the Cantor set.

EXAMPLE 11.2.2:

Show that the Cantor set C consists of all points $c \in [0,1]$ that have no 1's in their base-3 expansion.

Solution: The idea of expanding numbers in different bases may be unfamiliar, unless you were one of those children who was taught “New Math” in elementary school. Now you finally get to see why base-3 is useful!

First let's remember how to write an arbitrary number $x \in [0,1]$ in base-3. We expand in powers of $1/3$: thus if $x = \frac{a_1}{3} + \frac{a_2}{3^2} + \frac{a_3}{3^3} + \dots$, then $x = .a_1a_2a_3\dots$ in base-3, where the digits a_n are 0, 1, or 2. This expansion has a nice geometric interpretation (Figure 11.2.2).

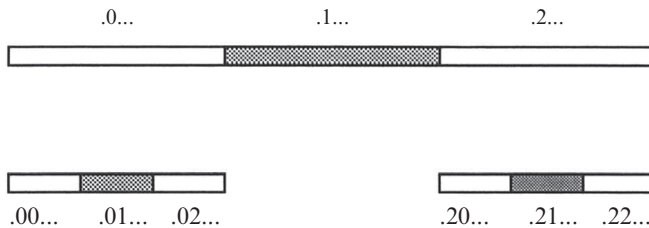


Figure 11.2.2

If we imagine that $[0,1]$ is divided into three equal pieces, then the first digit a_1 tells us whether x is in the left, middle, or right piece. For instance, all numbers with $a_1 = 0$ are in the left piece. (Ordinary base-10 works the same way, except that we divide $[0,1]$ into ten pieces instead of three.) The second digit a_2 provides more refined information: it tells us whether x is in the left, middle, or right third of a given piece. For instance, points of the form $x = .01\dots$ are in the middle part of the left third of $[0,1]$, as shown in Figure 11.2.2.

Now think about the base-3 expansion of points in the Cantor set C . We deleted the middle third of $[0,1]$ at the first stage of constructing C ; this removed all points whose first digit is 1. So those points can't be in C . The points left over (the only ones with a chance of ultimately being in C) must have 0 or 2 as their first digit. Similarly, points whose *second* digit is 1 were deleted at the next stage in the construction. By repeating this argument, we see that C consists of all points whose base-3 expansion contains no 1's, as claimed. ■

There's still a fussy point to be addressed. What about endpoints like $\frac{1}{3} = .1000\dots$? It's in the Cantor set, yet it has a 1 in its base-3 expansion. Does this contradict what we said above? No, because this point can also be written solely in terms of 0's and 2's, as follows: $\frac{1}{3} = .1000\dots = .02222\dots$. By this trick, each point in the Cantor set can be written such that no 1's appear in its base-3 expansion, as claimed.

Now for the payoff.

EXAMPLE 11.2.3:

Show that the Cantor set is uncountable.

Solution: This is just a rewrite of the Cantor diagonal argument of Example 11.1.4, so we'll be brief. Suppose there were a list $\{c_1, c_2, c_3, \dots\}$ of all points in C . To show that C is uncountable, we produce a point $\bar{c}$ that is in C but not on the list. Let c_{ij} denote the j th digit in the base-3 expansion of c_i . Define $\bar{c} = \bar{c}_{11}\bar{c}_{22}\dots$, where the overbar means we switch 0's and 2's: thus $\bar{c}_{nn} = 0$ if $c_{nn} = 2$ and $\bar{c}_{nn} = 2$ if $c_{nn} = 0$. Then $\bar{c}$ is in C , since it's written solely with 0's and 2's, but $\bar{c}$ is not on the list, since it differs from c_n in the n th digit. This contradicts the original assumption that the list is complete. Hence C is uncountable. ■

11.3 Dimension of Self-Similar Fractals

What is the “dimension” of a set of points? For familiar geometric objects, the answer is clear—lines and smooth curves are one-dimensional, planes and smooth surfaces are two-dimensional, solids are three-dimensional, and so on. If forced to give a definition, we could say that *the dimension is the minimum number of coordinates needed to describe every point in the set*. For instance, a smooth curve is one-dimensional because every point on it is determined by one number, the arc length from some fixed reference point on the curve.

But when we try to apply this definition to fractals, we quickly run into paradoxes. Consider the **von Koch curve**, defined recursively in Figure 11.3.1.

We start with a line segment S_0 . To generate S_1 , we delete the middle third of S_0 and replace it with the other two sides of an equilateral triangle. Subsequent stages are generated recursively by the same rule: S_n is obtained by replacing the middle third of each line segment in S_{n-1} by the other two sides of an equilateral triangle. The limiting set $K = S_\infty$ is the von Koch curve.

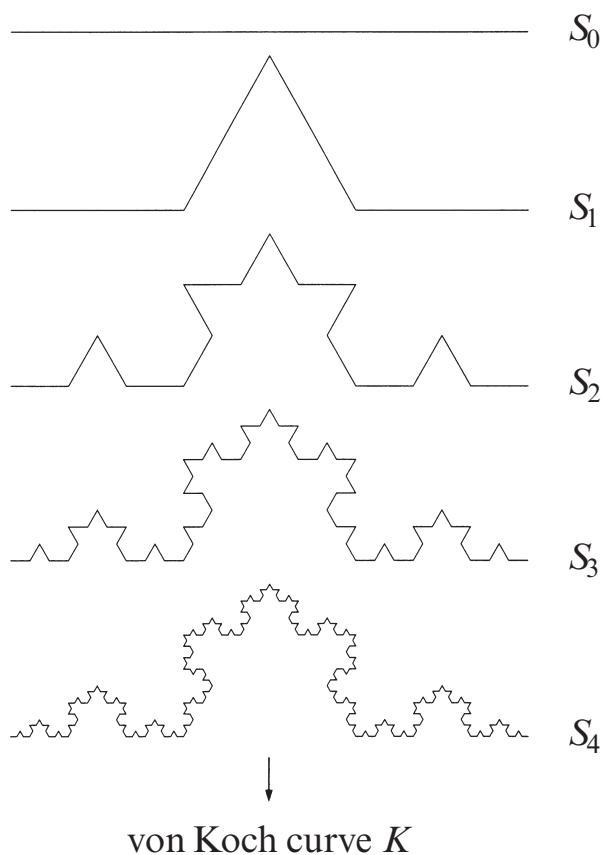


Figure 11.3.1

A Paradox

What is the dimension of the von Koch curve? Since it's a curve, you might be tempted to say it's one-dimensional. But the trouble is that K has *infinite arc length*! To see this, observe that if the length of S_0 is L_0 , then the length of S_1 is $L_1 = \frac{4}{3}L_0$, because S_1 contains four segments, each of length $\frac{1}{3}L_0$. The length increases by a factor of $\frac{4}{3}$ at each stage of the construction, so $L_n = (\frac{4}{3})^n L_0 \rightarrow \infty$ as $n \rightarrow \infty$.

Moreover, the arc length between *any* two points on K is infinite, by similar reasoning. Hence points on K aren't determined by their arc length from a particular point, because every point is infinitely far from every other!

This suggests that K is more than one-dimensional. But would we really want to say that K is two-dimensional? It certainly doesn't seem to have any "area." So the dimension should be *between* 1 and 2, whatever that means.

With this paradox as motivation, we now consider some improved notions of dimension that can cope with fractals.

Similarity Dimension

The simplest fractals are self-similar, i.e., they are made of scaled-down copies of themselves, all the way down to arbitrarily small scales. The dimension of such fractals can be defined by extending an elementary observation about *classical* self-similar sets like line segments, squares, or cubes. For instance, consider the square region shown in Figure 11.3.2.

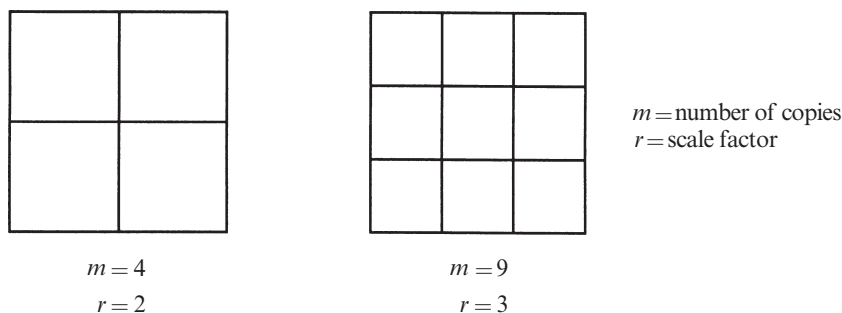


Figure 11.3.2

If we shrink the square by a factor of 2 in each direction, it takes four of the small squares to equal the whole. Or if we scale the original square down by a factor of 3, then nine small squares are required. In general, if we reduce the linear dimensions of the square region by a factor of r , it takes r^2 of the smaller squares to equal the original.

Now suppose we play the same game with a solid cube. The results are different: if we scale the cube down by a factor of 2, it takes eight of the smaller cubes to make up the original. In general, if the cube is scaled down by r , we need r^3 of the smaller cubes to make up the larger one.

The exponents 2 and 3 are no accident; they reflect the two-dimensionality of the square and the three-dimensionality of the cube. This connection between dimensions and exponents suggests the following definition. Suppose that a self-similar set is composed of m copies of itself scaled down by a factor of r . Then the **similarity dimension** d is the exponent defined by $m = r^d$, or equivalently,

$$d = \frac{\ln m}{\ln r}.$$

This formula is easy to use, since m and r are usually clear from inspection.

EXAMPLE 11.3.1:

Find the similarity dimension of the Cantor set C .

Solution: As shown in Figure 11.3.3, C is composed of two copies of itself, each scaled down by a factor of 3.

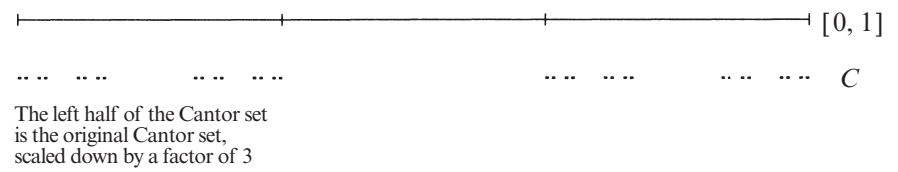


Figure 11.3.3

So $m = 2$ when $r = 3$. Therefore $d = \ln 2 / \ln 3 \approx 0.63$. ■

In the next example we confirm our earlier intuition that the von Koch curve should have a dimension between 1 and 2.

EXAMPLE 11.3.2:

Show that the von Koch curve has a similarity dimension of $\ln 4 / \ln 3 \approx 1.26$.

Solution: The curve is made up of four equal pieces, each of which is similar to the original curve but is scaled down by a factor of 3 in both directions. One of these pieces is indicated by the arrows in Figure 11.3.4.

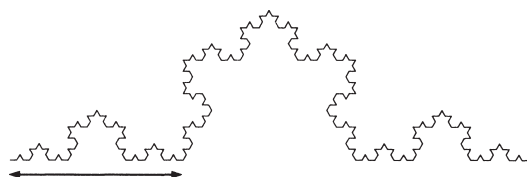


Figure 11.3.4

Hence $m = 4$ when $r = 3$, and therefore $d = \ln 4 / \ln 3$. ■

More General Cantor Sets

Other self-similar fractals can be generated by changing the recursive procedure. For instance, to obtain a new kind of Cantor set, divide an interval into five equal pieces, delete the second and fourth subintervals, and then repeat this process indefinitely (Figure 11.3.5).

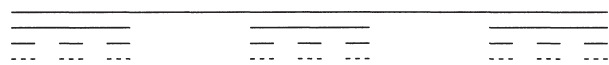


Figure 11.3.5

We call the limiting set the *even-fifths Cantor set*, since the even fifths are removed at each stage. (Similarly, the standard Cantor set of Section 11.2 is often called the *middle-thirds Cantor set*.)

EXAMPLE 11.3.3:

Find the similarity dimension of the even-fifths Cantor set.

Solution: Let the original interval be denoted S_0 , and let S_n denote the n th stage of the construction. If we scale S_n down by a factor of five, we get one third of the set S_{n+1} . Now setting $n = \infty$, we see that the even-fifths Cantor set S_∞ is made of three copies of itself, shrunk by a factor of 5. Hence $m = 3$ when $r = 5$, and so $d = \ln 3 / \ln 5$. ■

There are so many different Cantor-like sets that mathematicians have abstracted their essence in the following definition. A closed set S is called a **topological Cantor set** if it satisfies the following properties:

1. S is “totally disconnected.” This means that S contains no connected subsets (other than single points). In this sense, all points in S are separated from each other. For the middle-thirds Cantor set and other subsets of the real line, this condition simply says that S contains no intervals.
2. On the other hand, S contains no “isolated points.” This means that every point in S has a neighbor arbitrarily close by—given any point $p \in S$ and any small distance $\varepsilon > 0$, there is some other point $q \in S$ within a distance ε of p .

The paradoxical aspects of Cantor sets arise because the first property says that points in S are spread apart, whereas the second property says they’re packed together! In Exercise 11.3.6, you’re asked to check that the middle-thirds Cantor set satisfies both properties.

Notice that the definition says nothing about self-similarity or dimension. These notions are geometric rather than topological; they depend on concepts of distance, volume, and so on, which are too rigid for some purposes. Topological features are more robust than geometric ones. For instance, if we continuously deform a self-similar Cantor set, we can easily destroy its self-similarity but properties 1 and 2 will persist. When we study strange attractors in Chapter 12, we’ll see that the cross sections of strange attractors are often topological Cantor sets, although they are not necessarily self-similar.

11.4 Box Dimension

To deal with fractals that are not self-similar, we need to generalize our notion of dimension still further. Various definitions have been proposed; see Falconer (1990) for a lucid discussion. All the definitions share the idea of “measurement at a scale ε ”—roughly speaking, we measure the set in a way that ignores irregularities of size less than ε , and then study how the measurements vary as $\varepsilon \rightarrow 0$.

Definition of Box Dimension

One kind of measurement involves covering the set with boxes of size ε (Figure 11.4.1).

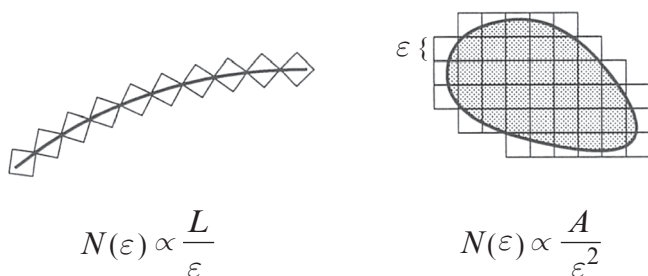


Figure 11.4.1

Let S be a subset of D -dimensional Euclidean space, and let $N(\varepsilon)$ be the minimum number of D -dimensional cubes of side ε needed to cover S . How does $N(\varepsilon)$ depend on ε ? To get some intuition, consider the classical sets shown in Figure 11.4.1. For a smooth curve of length L , $N(\varepsilon) \propto L/\varepsilon$; for a planar region of area A bounded by a smooth curve, $N(\varepsilon) \propto A/\varepsilon^2$. The key observation is that the dimension of the set equals the exponent d in the **power law** $N(\varepsilon) \propto 1/\varepsilon^d$.

This power law also holds for most fractal sets S , except that d is no longer an integer. By analogy with the classical case, we interpret d as a dimension, usually called the *capacity* or **box dimension** of S . An equivalent definition is

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)}, \text{ if the limit exists.}$$

EXAMPLE 11.4.1:

Find the box dimension of the Cantor set.

Solution: Recall that the Cantor set is covered by each of the sets S_n used in its construction (Figure 11.2.1). Each S_n consists of 2^n intervals of length $(1/3)^n$, so if we pick $\varepsilon = (1/3)^n$, we need all 2^n of these intervals to cover the Cantor set. Hence $N = 2^n$ when $\varepsilon = (1/3)^n$. Since $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$, we find